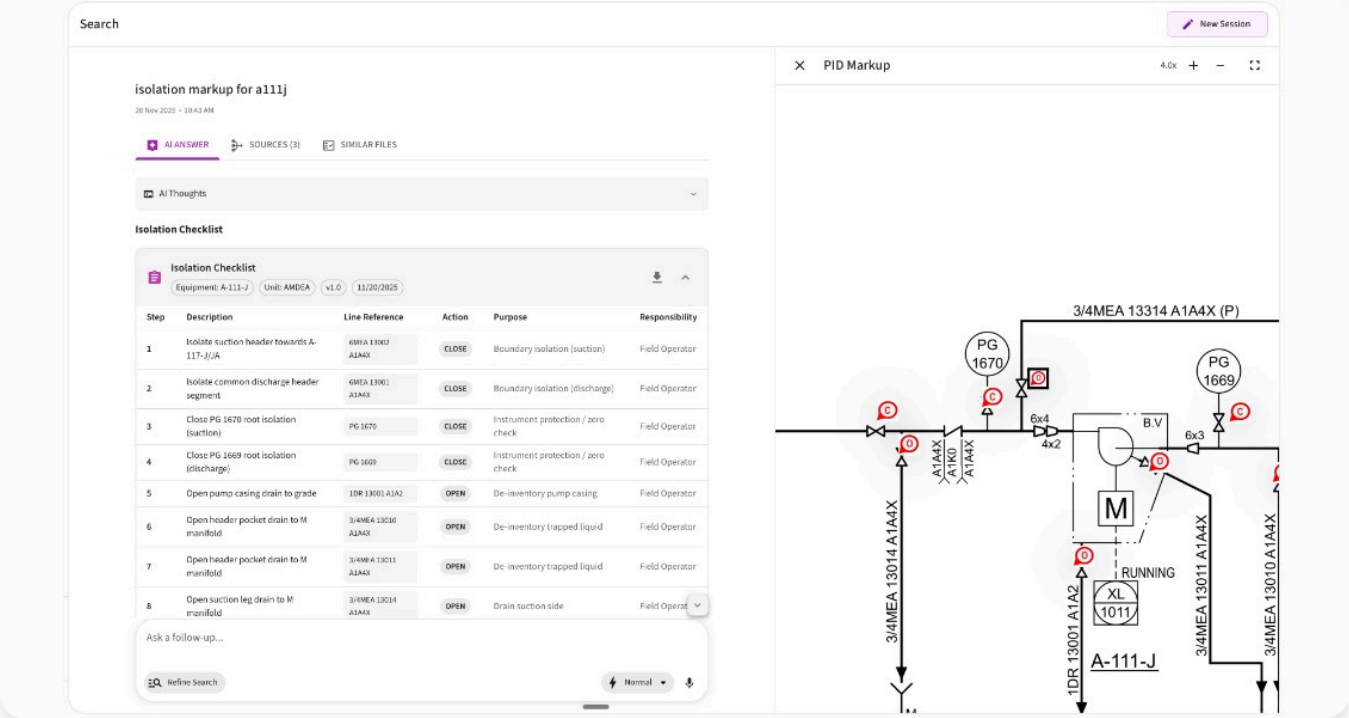
raven

ISOLATION PLANNING NOW

Supports LOTO / energy isolation requirements · OSHA 29 CFR 1910.147 · OISD-STD-105

Marked-up P&IDs and reviewable isolation plan checklists, generated from your plant documents.

Raven assists isolation planning inside your existing permit workflow. It marks up the P&ID, generates a cited isolation plan checklist from plant documents, and routes the draft through built-in review and approval so teams work from a more consistent starting point across shifts.

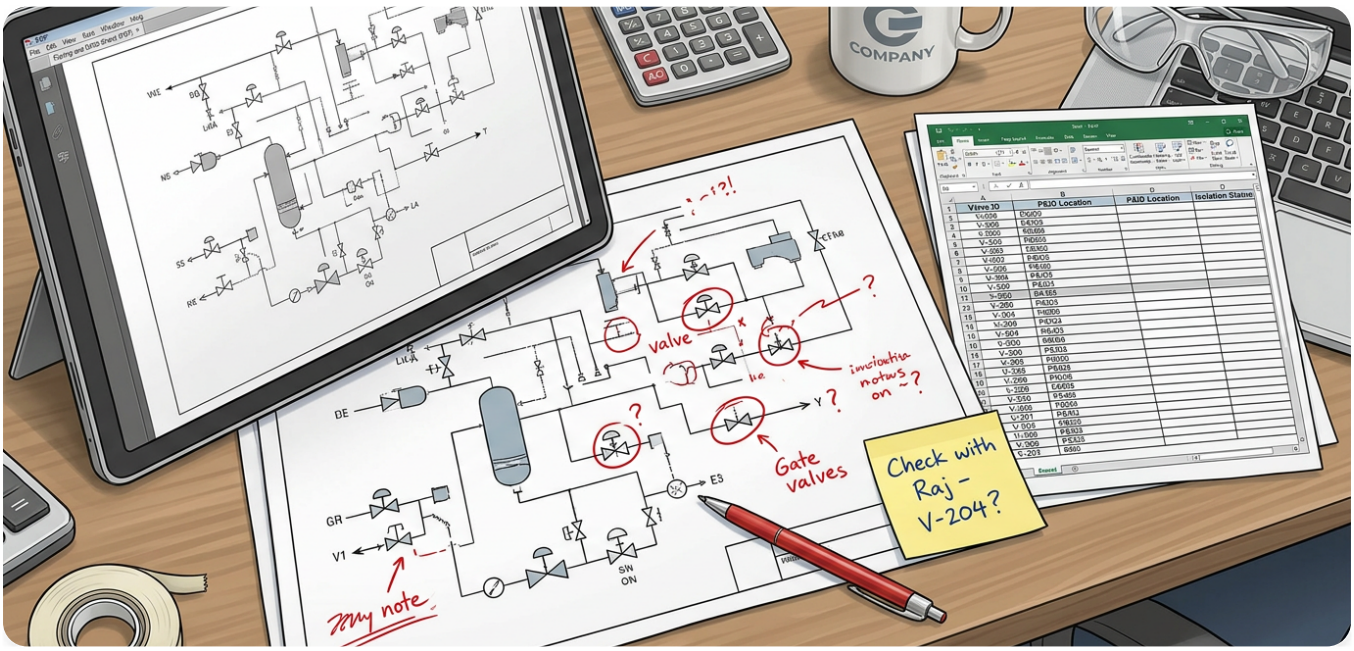


Is this relevant for your plant?

- Isolation pack preparation depends on one or two experienced people
- You've had permit delays or near-misses from missed isolation points
- P&IDs are referenced manually (paper or flat PDF) during permit prep
- Isolation quality noticeably drops during turnarounds or night shifts
- Knowledge transfer to newer permit coordinators is a concern

THE RISK TODAY

Isolation planning still depends heavily on the people who know the plant best. Experienced permit coordinators cross-reference P&IDs, SOPs, and their own knowledge to identify isolation points and sequence the work. But when that person is on leave, or it's a night shift, or turnaround pressure is high, maintaining the same quality becomes harder. Manual cross-referencing is where small gaps can enter, and in isolation work, small gaps carry real consequences.



How Raven changes this

EXAMPLE WORKFLOW

A permit coordinator opens the P&ID for the affected unit, selects the equipment to be worked on, reviews Raven's marked-up isolation boundary, checks the generated isolation plan checklist, then routes it through approval with notifications to the right reviewers.

1. **Select**

   Open the interactive P&ID and choose the equipment or section that needs to be isolated.

2. **Mark up**

   Raven marks up the likely isolation boundary on the drawing, including valves, vents, drains, and relevant energy sources.

3. **Generate**

   It turns that markup into an isolation plan checklist with sequence, precautions, and cited plant instructions.

4. **Review**

   The coordinator checks the markup and checklist, makes adjustments, and submits the draft into the approval flow.

5. **Approve**

   Reviewers are notified inside the workflow, approve the checklist, and the final isolation plan checklist is ready for permit attachment.

What you get

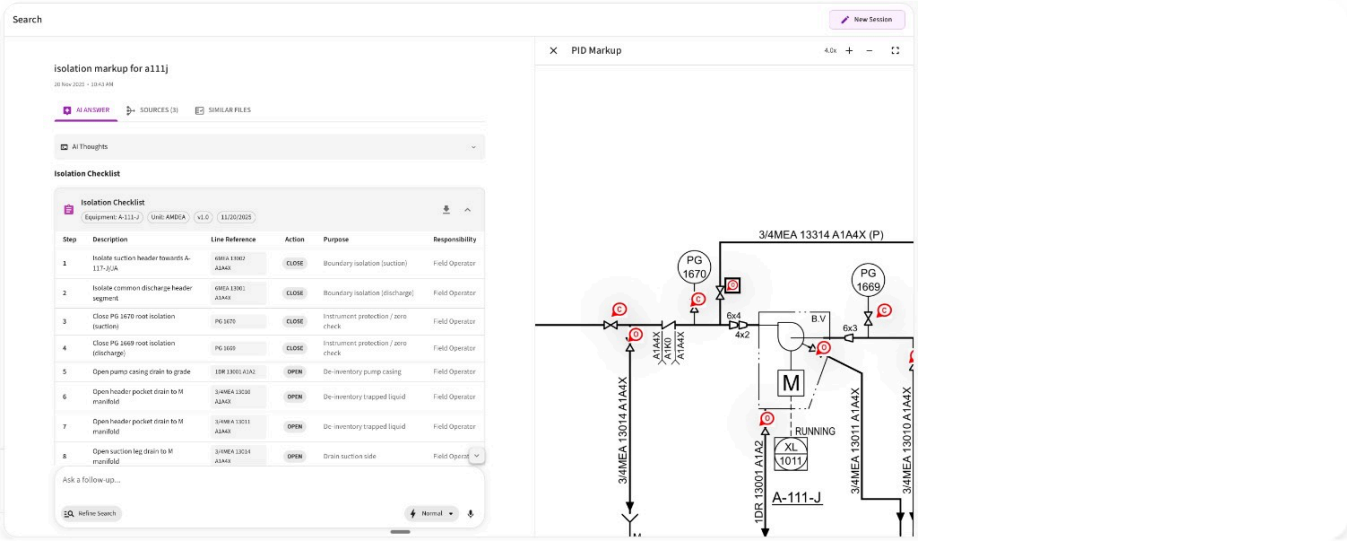
- Suggested isolation boundary marked up directly on the interactive P&ID
- Valve references with tag numbers and required positions
- Built-in approval workflow with reviewer notifications and sign-off
- Isolation plan checklist with step sequence and precautions
- Referenced SOPs and manual sections for each step
- Printable/downloadable pack — ready to attach to the permit

REVIEWABLE BY DESIGN

Each isolation plan checklist stays tied back to the marked-up drawing and plant documents, so reviewers can verify what was pulled, comment, and adjust before issue.

BUILT FOR

Permit Coordinator | Shift Supervisor | Ops Manager | Maintenance Planner | Plant Manager



Safety impact

Consistency

Packs start from the same P&ID and document workflow, not a different memory process each shift.

Reviewability

Reviewers can see the marked-up P&ID, the generated isolation plan checklist, and the cited basis for each step before approval.

Resilience

Less dependence on one or two experienced coordinators during leave, shift changes, or turnaround pressure.

KPIS THIS IMPROVES

- **Permit rework and review time** — reviewers start from a marked-up P&ID and referenced isolation plan checklist instead of rebuilding the logic manually
- **Permit preparation consistency** — more uniform pack quality across shifts and experience levels
- **Permit cycle time** — faster preparation helps permits move on schedule instead of being rushed late
- **Approval turnaround** — notifications and built-in routing help move reviews forward faster
- **Isolation planning resilience** — less disruption when experienced coordinators are unavailable

> *At Indorama Eleme, Raven is used to mark up P&IDs and generate referenced isolation plan checklists from plant documents, giving teams a more consistent starting point for review and approval across shifts and coordinators.*
>
> **— Indorama Eleme Petrochemicals, Nigeria**

PILOT SCOPE

See it on your plant's data

90-day pilot. One unit. Real documents. Measurable outcomes.

90-DAY PILOT

One workflow, one unit. Define scope, inputs, outputs, success metrics. Direct support from Raven's engineering team throughout.

CONTACT

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